



Apoorv Agnihotri

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Education 🏛️

10.2023 - 09.2026

Eberhard Karls University of Tübingen

MSC. MACHINE LEARNING (Grade: 1.85)

07.2016 - 07.2020

Indian Institute of Technology (IIT) Gandhinagar

BTECH. COMPUTER SCIENCE AND ENGINEERING (GPA: 8.25/10)

Research Experience 🧐

10.2025 - 07.2026

Robust Machine Learning Group, ELLIS Institute Tübingen

MASTER'S THESIS

Advisors: Dr. Wieland Brendel and Prof. Matthias Bethge

- **Thesis:** Evaluating AI-Supported Planning for Differentiated Mathematics Homework
- Designed and conducted a two-phase mixed-methods study combining teacher interviews with a controlled comparison of manual and AI-assisted planning platforms
- Found that AI assistance supported ideation and task discovery but with increased complexity and repair work

05.2019 - 05.2020

Sustainability Lab, IIT Gandhinagar

UNDERGRADUATE RESEARCHER

Advisor: Prof. Nipun Batra

- Authored an expository article on Bayesian Optimization (Distill 2020)
- Created Polire and Vayu for computational sustainability research (Poster at SenSys 2020)
- Applied active learning for optimal sensor placement (CoDS COMAD 2020, ICML Workshop 2020)

Industry Experience ⚙️

04.2022 - 06.2023

Rephrase AI (acquired by Adobe), Bangalore

DEEP LEARNING RESEARCHER

- Experimented with Conditional VAE-based voice cloning for zero-shot speaker adaptation with under 10s target audio
- Leveraged self-supervised speech encoders (hubert-soft) for voice content disentanglement, improving lip-sync
- Integrated pytorch-lightning and Hydra for reproducible ML experiments, establishing team best practices

06.2020 - 03.2022

Wadhvani AI, Delhi

ASSOCIATE ML SCIENTIST I

- Developed a medical document interpretation prototype employing classical computer vision techniques
- Deployed SOTA object detection models (SSD) on Android phones with pruning and on-device inference
- Investigated adversarial robustness via sample rejection for agricultural imagery (Workshop ICLR 2023)
- Won 1M INR at Google's AI for Agriculture Hackathon

Publications

2026	PREPRINT <i>arXiv:2605.28508</i>	A. Pant*, K. Shah*, A. Agnihotri* , S. Nikam, P. Balraj, and N. Jain	
2024	JOURNAL <i>J. Lightwave Technology</i>	A. Varri, F. Brückeroth-Plückelmann, J. Dijkstra, D. Wendland, R. Bankwitz, A. Agnihotri , and W. H. P. Pernice	
2023	WORKSHOP <i>PML4DC at ICLR</i>	J. White, C. Agrawal, A. Ojha, A. Agnihotri , M. Sharma, and J. Doshi	 
2022	WORKSHOP <i>NeurIPS Workshop</i>	J. White, P. Madaan, N. Shenoy, A. Agnihotri , M. Sharma, and J. Doshi	
2020	JOURNAL <i>Distill</i>	A. Agnihotri and N. Batra	 
2020	CONFERENCE <i>CoDS-COMAD YRS</i>	S. D. Narayanan, A. Agnihotri , and N. Batra	 
2020	POSTER <i>ACM SenSys</i>	S. D. Narayanan*, Z. B. Patel*, A. Agnihotri , and N. Batra	 
		Benchmarking AI for low-resource contexts: Thinking beyond leader-boards	
		Noise-Resilient Photonic Analog Neural Networks	
		BOLLWM: A Real-World Dataset for Bollworm Pest Monitoring from Cotton Fields in India	
		A Case for Rejection in Low Resource ML Deployment	
		Exploring Bayesian Optimization	
		Active Learning for Air Quality Station Location Recommendation	
		A Toolkit for Spatial Interpolation and Sensor Placement	

Additional Experience

05.2025 - 07.2025	Cyber Valley AI Incubator, MPI-IS, Tübingen PARTICIPANT	Completed an AI venture program spanning problem framing, prototyping, user validation, and market strategy
08.2024 - 05.2025	Mercedes-Benz, Böblingen AI RESEARCH - WORKING STUDENT	Experimented with neuromorphic image denoising (Locally Competitive Algorithm), evaluated trade-offs vs deep learning
07.2023 - 01.2024	Artpark, Indian Institute of Science, Bangalore ML ENGINEER CONSULTANT	Built RAG prototype on LLMs for pregnancy care knowledge dissemination to nursing mid-wives in rural India
05.2018 - 07.2018	NVIDIA, Bangalore ACCELERATED HPC & ML INTERN	Implemented GPU-optimized Kalman filters in rapids.ai data science framework using CUDA

Teaching & Mentorship

2022	Guest lecture on Diffusion Models in Probabilistic ML course @ IITGN	
2020 - 2023	Research seminars at Wadhvani AI and Rephrase AI	
2019	Teaching Assistant for Machine Learning (TEQIP) @ IITGN · Upskilling course for local college professors	
2018	PyData Talk on rapids.ai @ PyData, Gandhinagar	

Languages

- German (B1) · English (C2) · Hindi (Native)